

Changwoo Bae

Fluid dynamicist & Physicist

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Education

- PhD** Institut Lumière Matière, Université de Lyon, PhD in Physics, Material – Lyon, France 2021 – 2024
- Thesis: ‘Nanofluidics with Soap Bubbles and Surfactants’
 - Advisor: Dr. Anne-Laure Biance
 - Investigated the motion of an air bubble surrounded by ionic surfactants under an electric field.
 - Derived a physical model for the bubble motion.
 - Studied colloid filtration/seiving using bubble-in-capillary configuration.
 - Investigated "electronic" surfactants’ pi-stacking signature at water/air interface using Langmuir Trough and Ellipsometer combined setup.
 - Built a setup to measure conductivity of pi-stacked surfactants and graphene flakes at water/air interface.
- M.E.** KyungHee University, M.E. in Mechanical Engineering – Yongin, South Korea 2018 – 2020
- Thesis: ‘Penetration Dynamics of Water Droplet on Janus Mesh’
 - Advisor: Prof. Choongyeop Lee
 - Investigated on drop impact dynamics on engineered surfaces such as superhydrophobic(SHPo) surfaces, Lubricant-infused surfaces(LIS), Janus membrane, etc.
 - Developed models for penetration dynamics of water droplets on Janus membrane.
 - Studied a contact time reduction on a curved superhydrophobic surface during drop impact.
 - Demonstrated rebound and rotation phenomena on a inclined LIS.
- B.E.** KyungHee University, B.E. in Mechanical Engineering – Yongin, South Korea 2010 – 2018
- Advisor: Prof. Choongyeop Lee
 - Studied free energy harvesting from salt concentration gradient.
 - Demonstrated an ionic diode effect from asymmetric contact areas.

Experience

- SIMM, ESPCI Paris-PSL University**, Postdoctoral Researcher – Paris, France 2025 – present
- Investigating liquid-triboelectrification.
 - Investigating polymer adsorption dynamics.
- Institut Lumière Matière, Université de Lyon**, Postdoctoral Researcher – Lyon, France 2025 – 2025
- Extended Ph.D. research on electro-driven bubble transport with tunable surfactant concentration.
 - Investigated electronic interactions of oligothiophene-based surfactants at liquid/gas interfaces.
 - Mentored a master’s student on experimental techniques and data analysis.
- Inha University**, Research Fellow – Incheon, South Korea 2021 – 2021
- Characterized various properties of hydrogels.
 - Developed novel microfluidic devices for studying diffusioosmosis on superhydrophobic surfaces.
- KyungHee University**, Research Fellow – Yongin, Korea 2020 – 2020
- Studied diffusioosmosis/phoresis on superhydrophobic channel.

Publications

- Reversing the electro-driven bubble transport by tuning surfactant concentration** In preparation
Changwoo Bae, Menghua Zhao, Christophe Ybert, Anne-Laure Biance
- Electronic interactions of a quaterthiophene-based surfactant at the liquid/gas interface** 2025
Changwoo Bae, Kamatham Narayanaswamy, Hisham Idriss, Ludivine Poyac, Indraneel Sen, Sébastien Richeter, Sébastien Clément, Anne-Laure Biance, Samuel Albert, Oriane Bonhomme
10.1039/D5SM00122F (Soft Matter)

Molecular Assemblies of Amphiphilic Oligothiophenes at the Air–Water Interface	2025
Hisham Idriss, Samuel Albert, <i>Changwoo Bae</i> , Ludivine Poyac, Philippe Gerbier, Oriane Bonhomme, Joséphine Lai-Kee-Him, Aurélie Ancelin, Sébastien Richeter, Indraneel Sen, Karl Leo, Anne-Laure Biance, Sébastien Clément 10.1021/acs.langmuir.5c01129 (Langmuir)	
A bouncing and rotating drop after oblique impact on Lubricant-Impregnated surfaces	2024
<i>Changwoo Bae</i> , Young-Su Ko, Seungwon Shin, Choongyeop Lee 10.1063/5.0239361 (Physics of Fluids)	
Highly porous hydrogels for efficient solar water evaporation	2024
Akash Ranjan Pati, Young-Su Ko, <i>Changwoo Bae</i> , Inhee Choi, Yun Jung Heo, Choongyeop Lee 10.1039/D4SM00388H (Soft Matter)	
Water penetration dynamics through a Janus mesh during drop impact	2020
<i>Changwoo Bae</i> , Seungtae Oh, Jeonghoon Han, Youngsuk Nam, Choongyeop Lee 10.1039/D0SM00567C (Soft Matter)	
Contact time on curved superhydrophobic surfaces	2020
Jeonghoon Han, Wonjung Kim, <i>Changwoo Bae</i> , Dongwook Lee, Seungwon Shin, Choongyeop Lee, Youngsuk Nam 10.1103/PhysRevE.101.043108 (Physical Review E)	
High-efficiency power generation in hyper-saline environment using conventional nanoporous membrane	2019
Jeonghoon Han, <i>Changwoo Bae</i> , Songhwa Chae, Dukhyun Choi, Sangmin Lee, Youngsuk Nam, Choongyeop Lee 10.1016/j.electacta.2019.07.005 (Electrochimica Acta)	

Conference Talks

Charged Matter 2025	2025
15th European Foam Conference (EUFOAM)	2024
APS Division of Fluid Dynamics Meeting	2023
Journées Plenieres GDR MicroNanoFluidique	2023
Congres Français de Mécanique	2022
Journées de Physique Statistique	2022
The Korean BioChip Society	2021
The 11th National Congress of Fluids Engineering	2020
The Korean Society of Mechanical Engineering Fall Conference	2019
The Korean Society of Mechanical Engineering Spring Conference	2019
The 10th National Congress on Fluids Engineering	2018

Invited Talks

Sciences et Ingénierie de la Matière Molle, ESPCI	2025 Jul
Institut de Physique de Rennes (IPR)	2025 Jan

Skills

Experimental Techniques: High-speed imaging, Microfluidics, Surface coating, Tensiometry, Spectroscopy ellipsometry, Atomic Force Microscopy

Programming & Software: Python, LaTeX, Rhinoceros 3D, Adobe Illustrator, Inkscape

Fabrication: Soft lithography, Surface functionalization

Expertise: Hydrodynamics, Thermodynamics, Electrokinetics, Interfacial physics

Teaching & Mentoring

Mentor for three Master's students	2021 – 2024
Research assistant	2019 – 2020

Extracurricular Activities

Complex Motion in Fluids, Homerton College, Cambridge, UK	2023
International School of Soft Matter, Cargèse, France	2022

Professional Activities

- Reviewer for Physics of Fluids
- Reviewer for Microfluidics and Nanofluidics

Languages

English, Korean

Awards

Excellent Paper Award The Korean Society of Mechanical Engineers	2020
Academic Excellence Scholarship KyungHee University	2017
Silver Prize in Korean Physics Olympiad Korea_	2009